American Computer Science League

2020-2021 ● Contest 3: Shorts ● Elementary Division

1. Boolean Algebra

Determine whether this Boolean expression is TRUE or FALSE:

$(7 \geq 4 * 2)$ OR NOT $(5 - 3 < 2$ AND $1 + 3 \neq 5)$

A. TRUE
B. FALSE
C. Invalid expression
D. Not always the same

2. Boolean Algebra

How many ordered pair(s) make the following expression TRUE?

NOT (A AND NOT B) OR A

A. 0
B. 1
C. 2
D. 3
E. 4

3. Boolean Algebra

Identify the only ordered pair that makes both expressions TRUE.

~(A OR B) OR ~A ~(~A) AND A OR B

A. (0,0)
B. (0,1)
C. (1,0)
D. (1,1)
E. More than 1 of these

4. Boolean Algebra

Simplify the following Boolean expression to the fewest operators:

~A(~AB + A~B)

A. ~A
B. ~AB
C. A~B
D. ~B
E. ~A+B

5. Boolean Algebra

Which of the following is NOT equivalent to the other three?

A. ~A + ~A~B B. ~(A + B)

C. ~A~A~B D. ~AA + ~A~B

A. ~A + ~A~B
B. ~(A + B)
C. ~A~A~B
D. ~AA + ~A~B
E. they are all equivalent

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1. Boolean Algebra

判断下述布尔表达式是真还是假：

$(7 \geq 4 * 2)$ OR NOT $(5 - 3 < 2$ AND $1 + 3 \neq 5)$

A. 表达式为真
B. 表达式为假
C. 非法表达式
D. 无法确定

2. Boolean Algebra

有多少个有序对能使下述表达式为真？

NOT (A AND NOT B) OR A

A. 0
B. 1
C. 2
D. 3
E. 4

3. Boolean Algebra

找出能使下述两个表达式均为真的唯一有序对。

~(A OR B) OR ~A ~(~A) AND A OR B

A. (0,0)
B. (0,1)
C. (1,0)
D. (1,1)
E. 不止一个有序对

4. Boolean Algebra

化简下述布尔表达式，使得其中包含的运算符数量最少：

~A(~AB + A~B)

A. ~A
B. ~AB
C. A~B
D. ~B
E. ~A+B

5. Boolean Algebra

下述哪个表达式和其他三个表达式是不等价的？

A. ~A + ~A~B B. ~(A + B)

C. ~A~A~B D. ~AA + ~A~B

A. ~A + ~A~B
B. ~(A + B)
C. ~A~A~B
D. ~AA + ~A~B
E. 四个表达式均是等价的